



# AFA Guidelines for Analyzing Duo-cone Seal Failures

Maintenance and  
Repair

Application

Component Life  
Management

Component  
Renewal (CRC)

MARC  
Management

|                                                     |   |
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## 1.0 Introduction

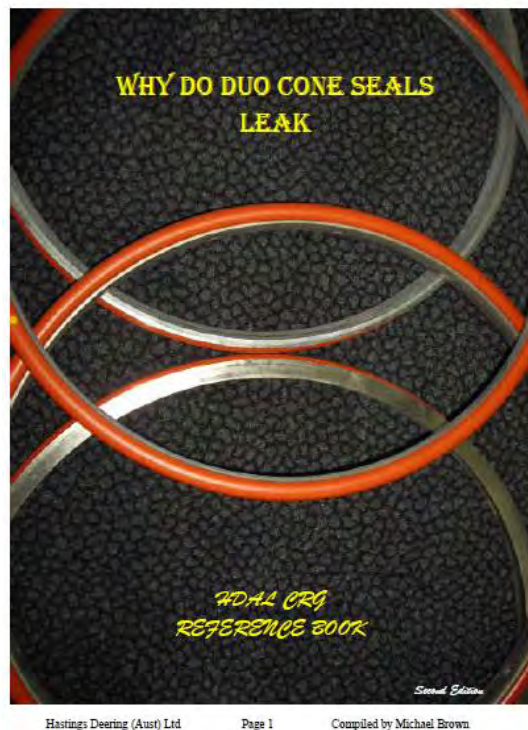
Numerous Caterpillar published Applied Failure Analysis (AFA) guidelines and training packages exist, however, most of the guidelines apply to metallurgical parts failures. There are few guidelines that discuss seal or other non-metal parts failure analysis.

Determining the root cause of any failure is important so that actions can be taken to avoid future repeat failures.

## 2.0 Best Practice Description

At this time, Caterpillar does not have an official guideline on analyzing duo-cone seal failures. Reuse and salvage guidelines do exist, but do not specifically cover the most common causes of failure.

Hastings Deering has written an internal guideline to assist their technicians and AFA staff with the analysis of duo-cone seal failures. The guideline is attached to this Best Practice.



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### 3.0 Implementation Steps

Since the document is not an official Caterpillar publication at this time, it is intended for *dealer use only*.

The principles contained in the guideline can be used to assist your AFA staff with duo-cone seal failure analysis and technician training.



FIG: 1g

View of an OHT toric ring showing a ridge formed by contact with the duo cone retainer lip.

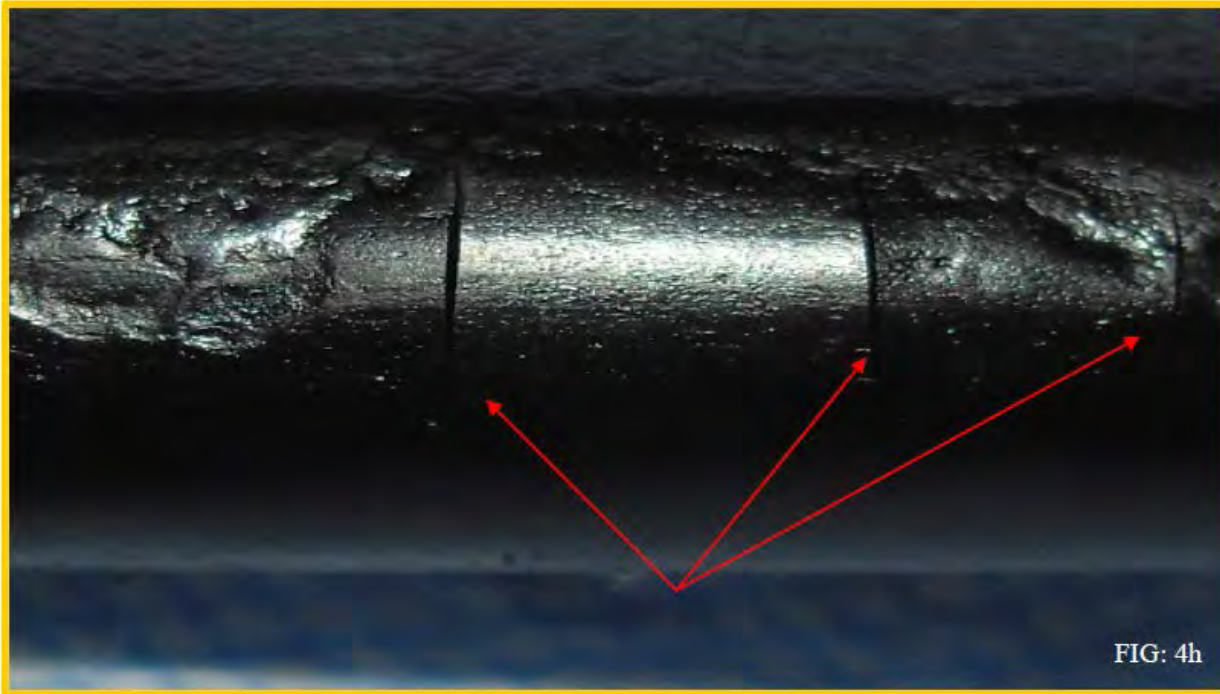
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View showing a toric ring with radial cracks. In this case caused by high temperatures and a long service life.

#### 4.0 Benefits

Use “root cause” information to take actions to prevent future repeat failures.  
Provide professional quality AFA reports.

#### 5.0 Resources Required

Staff time to review or incorporate into technician training.

#### 6.0 Supporting Attachments / References

- A GUIDE TO WHY DUO CONE SEALS LEAK2.pdf

#### 7.0 Related Best Practices

#### 8.0 Acknowledgements

Michael Brown – Hastings Deering

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